

Assessment 1 — Identification

Information Technology 10 (IT 10) • Arduino Family and Arduino SIK Components • 20 items • Answer key included

Learning Area	Information Technology 10 (TLE / ICT — Computer Systems & Electronics)	Test Type	Identification
Content / Topic	Arduino Family and Arduino SIK (SparkFun Inventor's Kit) Components	No. of Items	20
Date	_____	Rating	____ / 20

Table of Specification (TOS)

Content / Topic	Learning Competency	No. of Items	%	Cognitive Level		
				Remember	Understand	Apply
A. The Arduino Family (boards, microcontroller, IDE, sketch, program structure)	Identifies the members of the Arduino family and their core parts; describes the purpose of the Arduino IDE, sketch, <i>setup()</i> and <i>loop()</i> .	10	50%	6	3	1
B. Arduino SIK Components (input, output, and passive components of the Inventor's Kit)	Identifies SIK components (LED, resistor, potentiometer, photoresistor, temperature sensor, buzzer, motor, transistor, button, breadboard) and classifies each as input, output, or passive device.	10	50%	5	3	2
TOTAL		20	100%	11	6	3
Percentage per cognitive level				55%	30%	15%

Reference: Official Arduino documentation (arduino.cc) and the SparkFun Inventor's Kit (SIK) Guide. |

Directions: Write the correct term on the space provided. Spelling counts. (1 point each.)

Part I — The Arduino Family (Items 1-10)

1. It is the open-source electronics platform built on easy-to-use hardware (boards) and software (IDE).
2. The most common Arduino board, based on the ATmega328P microcontroller, widely used by beginners.
3. The Arduino board with 54 digital I/O pins and 16 analog inputs, designed for projects that need many connections.
4. The round, sewable Arduino board designed for wearable and e-textile projects.
5. The chip on the Arduino board that stores and executes the uploaded program (e.g., ATmega328P on the Uno).
6. The official software application used to write, compile, and upload programs to an Arduino board.

7. The name of a program written for the Arduino board (file extension *.ino*).
8. The function that runs once when the board is powered on or reset.
9. The function that runs over and over again continuously after *setup()* finishes.
10. The named connection points on an Arduino (DIGITAL, ANALOG IN, POWER) where components or jumper wires are attached.

Part II — Arduino SIK Components (Items 11-20)

1. The SIK component that gives off light when electric current flows through it in the correct direction.
2. The passive SIK component that limits the amount of current flowing in a circuit.
3. The SIK component with a rotating knob that acts as an adjustable voltage divider (variable resistor).
4. The SIK sensor whose resistance changes depending on the amount of light that hits it.
5. The SIK sensor used to measure temperature (commonly the TMP36).
6. The SIK component that converts electrical signals into sound.
7. The SIK component that converts electrical energy into rotary motion.
8. The semiconductor SIK component used as an electronic switch so that a small signal can control a high-current device such as a motor.
9. The simple SIK input device pressed by the user to open or close a circuit (momentary switch).
10. The solderless board where SIK circuits are built before connecting them to the Arduino.

Answer Key — Assessment 1

No.	Answer	No.	Answer
1	Arduino	11	Light-Emitting Diode (LED)
2	Arduino Uno	12	Resistor
3	Arduino Mega 2560	13	Potentiometer
4	Arduino LilyPad	14	Photoresistor / Light-Dependent Resistor (LDR)
5	Microcontroller	15	Temperature sensor (TMP36)
6	Arduino IDE	16	Piezo buzzer / Buzzer
7	Sketch	17	DC motor / Motor
8	setup()	18	Transistor
9	loop()	19	Push button / Tactile button
10	Pins (I/O pins)	20	Breadboard

Scoring guide: 1 point per correct answer; 20 points total. Accept equivalent answers where indicated (e.g., "LDR" for photoresistor, "buzzer" for piezo buzzer). Minor spelling errors that do not change the term may be accepted at the teacher's discretion.

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